

Yewei Huang, Ph.D.

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RESEARCH INTEREST

My main research interest is the challenges associated with **robot perception, navigation** and **decision-making** problems, particularly in the context of **multi-robot systems**. By deploying these algorithms in real-world environments on **physical robot systems** with **limited communication bandwidth**, I aim to enable fully autonomous robots to support the long-term maintenance of the underwater infrastructures and effectively monitor the complex offshore hydrodynamic and environmental conditions.

I'm interested in marine robot, multi-robot systems, SLAM, exploration, inter-robot loop closure, object-based data association, factor graph optimization, distributed multi-robot SLAM, planning under uncertainty, Gaussian process motion planning, expectation-maximization, learning-based decision making, multi-sensor fusion.

EDUCATION

<i>Doctor of Philosophy</i> , Mechanical Engineering Stevens Institute of Technology, NJ	08/2019 - 05/2025
<i>Master of Science</i> , Surveying Engineering Tongji University, Shanghai, China	09/2016 - 03/2019
<i>Bachelor of Science</i> , Geographic Information System Tongji University, Shanghai, China	09/2012 - 06/2016

POSITIONS

<i>Post-doctoral Research Fellow</i> , Dartmouth Reality and Robotics Lab Dartmouth College, Hanover, NH	07/2025 - present
<i>Summer Research Internship</i> , Vision and Robotics Laboratory SRI International, Princeton, NJ	05/2024 - 08/2024
<i>Graduate Student Research Assistant</i> , Robust Field Autonomy Laboratory Stevens Institute of Technology, Hoboken, NJ	08/2019 - 05/2025

PROJECT 1 - DISTRIBUTED MULTI-ROBOT PERCEPTION AND NAVIGATION

Robust Field Autonomy Laboratory, Stevens Institute of Technology, NJ Advisor: Dr. Brendan Englot

[*Distributed Multi-Robot LiDAR SLAM*](#) Jan 2020 – Sep 2021

- First distributed multi-robot SLAM system using compact LiDAR descriptors for **efficient data communication**.
- A two-step global-and-local pose graph optimization algorithm for accuracy and scalability.
- Manuscript published in IEEE Robotics and Automation Letters.
- Funded by National Science Foundation (NSF).

[*Multi-Robot SLAM for Underwater Unmanned Vehicle \(UUV\)s*](#) September 2024 - March 2025

- First to introduce distributed graph matching for underwater robot teams to enable **efficient data communication**.
- Inter-robot data association **outlier detection** algorithm specific for multi-robot sonar SLAM.
- Manuscript published in IROS'25.
- Funded by National Science Foundation (NSF) and Office of Naval Research (ONR).

Multi-Robot Autonomous Exploration

Jun 2023 - May 2024

- Multi-robot exploration while accounting for **localization uncertainty**.
- Reinforcement learning-based multi-robot exploration with Graph Neural Networks.
- Manuscript published in ICRA'24.
- Funded by Office of Naval Research (ONR).

PROJECT 2 - SYMBIOTIC DESIGN FOR CYBER PHYSICAL SYSTEMS CHALLENGE

Robust Field Autonomy Laboratory, Stevens Institute of Technology, NJ Advisor: Dr. Brendan Englot

Gaussian Process Motion Planning for UAVs

Aug 2022 - May 2023

- First **3D Motion planning algorithm for underwater vehicle** near seafloor.
- Motion planning algorithm with **oceanic current** considered.
- Manuscript published in IEEE Robotics and Automation Letters.
- Funded by Defense Advanced Research Projects Agency (DARPA).

PROJECT 3 - WALK-THROUGH RENDERING FROM IMAGES OF VARYING ALTITUDE

Robust Field Autonomy Laboratory, Stevens Institute of Technology, NJ Advisor: Dr. Brendan Englot

Structure From Motion(SFM) with Varying Altitude Images

Jan 2024 - Apr 2024

- First **geo-referenced** Structure-from-Motion pipeline using satellite images.
- Bundle Adjustment with geographical information encoded.
- Manuscript published in IROS'25.
- Funded by Intelligence Advanced Research Projects Activity.

PROJECT 4 - SUWAC SHARED UNDERSTANDING FOR WIDE-AREA HUMAN-ROBOT COLLABORATION

Vision and Robotics Laboratory, SRI International, NJ

Advisor: Dr. Han-Pang Chiu

Multi-robot Scene Graph SLAM

May 2024 - Aug 2024

- Distributed multi-robot SLAM with object matching for **efficient data communication**.
- Hierarchical scene graph for humans and robots to understand.
- Manuscript submitted.
- Licenced to Avsr AI.

PROJECT 5 - 3D SONAR PERCEPTION AND NAVIGATION

Reality and Robotics Lab, Dartmouth College, NH

Advisor: Dr. Alberto Quattrini Li

SLAM with 3D Sonar

July 2025 - present

- First underwater SLAM algorithm using **3D sonar**.
- First **3D sonar-camera calibration** for underwater environments.
- Manuscript accepted in ICRA'26.
- Funded by National Science Foundation (NSF) .

Marine Autonomous Exploration

Jan 2026 - present

- Exploration balancing **exploration and exploitation** in challenging marine environment.
- Uncertainty aware path planning in coastal environments.
- Manuscript submitted.
- Funded by National Science Foundation (NSF) .

AWARDS AND HONORS

- Academic Year 2019-2020 Stevens Provost Doctoral Fellowship
- Academic Year 2024-2025 Stevens Excellence Doctoral Fellowship
- Maryland Robotics Center: Future Leaders in Robotics and AI of 2025
- Academic Year 2024-2025 Paul Kaplan Award for Distinguished Doctoral Work upon Graduation
- Honorable Mention for the inaugural IEEE RAS Women in Engineering Best PhD Award in Robotics and Automation (2026)

PUBLICATIONS

1. Y. Li, **Y. Huang**, W. Gaozhang, A. Quattrini Li, B. Englot, and Y. Liu “[Variable-Resolution Virtual Maps for Autonomous Exploration with Unmanned Surface Vehicles \(USVs\)](#)” *Submitted, under review, 2026.*
2. J. McConnell, **Y. Huang**, T. Morris, J. Doughty, and D. Moynihan “Multi-session SLAM for Imaging Sonar Equipped Underwater Vehicles Using Semantic Scene Graphs” *Submitted, under review, 2026.*
3. C. Burgul, **Y. Huang**, M. Chatzisprou, I. Rekleitis, A. Quattrini Li, and M. Xanthidis “[Underwater dense mapping with the first compact 3d sonar](#)” *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
4. **Y. Huang**, T. Shan, A. Rajvanshi, N. Chowdhury, Y. Li, B. Englot, and H.-P. Chiu “[SGM-SLAM: Scene Graph Matching for Data-Efficient Distributed SLAM](#)” *Submitted, under review, 2025.*
5. **Y. Huang**, J. McConnell, X. Lin, and B. Englot “[DRACo-SLAM2: Distributed Robust Acoustic Communication-efficient SLAM for Imaging Sonar Equipped Underwater Robot Teams with Object Graph Matching](#)” *Submitted to IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
6. Y. Li, **Y. Huang**, W. Gaozhang, and Y. Liu “[Autonomous Exploration with Virtual Map using Unmanned Surface Vehicles](#)” *Submitted to IEEE International Conference on Automation and Computing (ICAC)*, 2025.
7. Y. Li, **Y. Huang**, B. Gaudel, H. Jafarnejadsani, and B. Englot “[CVD-SfM: A Cross-View Deep Front-end Structure-from-Motion System for Sparse Localization in Multi-Altitude Scenes](#)” *Submitted to IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
8. **Y. Huang**, X. Lin, and B. Englot “[Multi-Robot Autonomous Exploration and Mapping Under Localization Uncertainty with Expectation-Maximization](#)” *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 7236–7242, 2024.
9. **Y. Huang**, X. Lin, M. Hernandez-Rocha, S. Narain, K. Pochiraju and B. Englot “[Mission-oriented Gaussian Process Motion Planning for UUVs over Complex Seafloor Terrain and Current Flows](#)” *IEEE Robotics and Automation Letters (RA-L)*,9(2):1780-1787, 2024.
10. X. Lin, P. Szenher, **Y. Huang**, and B. Englot “[Distributional Reinforcement Learning Based Integrated Decision Making and Control for Autonomous Surface Vehicles](#)” *IEEE Robotics and Automation Letters (RA-L)*, 2024.

11. X. Lin, **Y. Huang**, F. Chen, and B. Englot “Decentralized Multi-Robot Navigation for Autonomous Surface Vehicles with Distributional Reinforcement Learning” *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
12. K. Doherty, A. Papalia, **Y. Huang**, D. Rosen, B. Englot, and J. Leonard “MAC: Maximizing Algebraic Connectivity for Graph Sparsification” *arXiv e-prints*, arXiv-2403, 2024.
13. X. Lin, **Y. Huang**, D. Sun, T.-Y. Lin, B. Englot, R. M. Eustice, and M. Ghaffari “A Robust Keyframe-Based Visual SLAM for RGB-D Cameras in Challenging Scenarios” *IEEE Access*, 11, 97239–97249, 2023.
14. J. Wang, F. Chen, **Y. Huang**, J. McConnell, T. Shan, and B. Englot “Virtual maps for autonomous exploration of cluttered underwater environments” *IEEE Journal of Oceanic Engineering*, 47(4):916–935, 2022.
15. J. McConnell, **Y. Huang**, P. Szenher, I. Collado-Gonzalez and B. Englot “DRACo-SLAM: Distributed Robust Acoustic Communication-efficient SLAM for Imaging Sonar Equipped Underwater Robot Teams” *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 8457–8464, 2022.
16. F. Chen, P. Szenher, **Y. Huang**, J. Wang, T. Shan, B. Shi and B. Englot “Zero-shot reinforcement learning on graphs for autonomous exploration under uncertainty” *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 5193–5199, 2021.
17. **Y. Huang**, T. Shan, F. Chen and B. Englot “DiSCo-SLAM: Distributed scan context-enabled multi-robot lidar slam with two-stage global-local graph optimization” *IEEE Robotics and Automation Letters (RA-L)*, 7(2):1150–1157, 2021.
18. F. Chen, J. Martin, **Y. Huang**, J. Wang, and B. Englot “Autonomous exploration under uncertainty via deep reinforcement learning on graphs” *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 6140–6147, 2020.
19. J. Zhao, **Y. Huang**, X. He, S. Zhang, C. Ye, T. Feng, and L. Xiong “Visual semantic landmark-based robust mapping and localization for autonomous indoor parking” *Sensors*, 19(1):161, 2019.
20. **Y. Huang**, J. Zhao, X. He, S. Zhang, and T. Feng “Vision-based semantic mapping and localization for autonomous indoor parking” *Proceedings of the IEEE Intelligent Vehicles Symposium (IV)*, 636–641, 2018.
21. **Y. Huang**, H. Wang, K. Zhan, J. Zhao, P. Gui, and T. Feng “Image-based localization for indoor environment using mobile phone” *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, 40:211–215, 2015.

MENTORING

I have mentored the following undergraduate and master’s students with my doctoral advisor, Prof. Brendan Englot:

- **Thomas Roff**. “Optimization of Utility Function Weights for Active Autonomous Exploration and Mapping With Multi-robot Systems” *Master’s thesis*, Stevens Institute of Technology, 2024.
- **Sneha Banda, Jesse Knuckles**. “Modular Sensor Fusion Package for Localization” *Undergraduate Senior Design*, sponsored by Kearfott Guidance and Navigation, Stevens Institute of Technology, 2025.

I have mentored the following doctoral student in collaboration with Prof. Yuanchang Liu from UCL:

- **Ye Li**. “Autonomous Exploration with Virtual Map using Unmanned Surface Vehicles” *The 30th International Conference on Automation and Computing (ICAC)*, 2025.

I have mentored the following doctoral student with my postdoctoral advisor, Prof. Alberto Quattrini Li, and Prof. Ioannis Rekleitis from University of Delaware:

- **Chinmay Burgul.** “Underwater dense mapping with the first compact 3d sonar” *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

SERVICES

Editorial Board

- Associate Editor, IEEE International Conference on Robotics and Automation (ICRA), 2026.

Reviewer

- IEEE Transactions on Robotics (T-RO).
- International Journal of Robotics Research (IJRR).
- IEEE Transactions on Field Robotics (T-FR).
- IEEE Robotics and Automation Letters (RA-L).
- Journal of Field Robotics (JFR).
- Elsevier Applied Ocean Research.
- Robotics: Science and Systems (RSS).
- IEEE International Conference on Robotics and Automation (ICRA).
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

TALKS

- 03/28/2025, Maryland Robotics Center (University of Maryland): Future Leaders in Robotics and AI Seminar
Data Efficient Localization and Mapping for Distributed Multi-Robot Teams in the Field
- 07/11/2025, University College London: Invited talk (virtual),
Distributed Robust Acoustic Communication-efficient SLAM for Imaging Sonar Equipped Underwater Robot Teams with Object Graph Matching
- 02/13/2026, School of Civil and Environmental, University of Connecticut: Environmental Engineering Seminar Talk
Reliable Navigation and Mapping in Uncertain Coastal Waters: Towards Trustworthy Marine Autonomy
- 04/27/2026, Stevens Institute of Technology: ME 656 Autonomous Navigation for Mobile Robots
Machine Learning for Visual SLAM from Scene Representation Perspective

SKILLS

C++, Python, ROS, reinforcement learning, computer vision, Matlab, LaTeX, Git.